

WEEK 2 (Praveen 2520030406)

2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int source, destination;
    char buffer[100];
    int bytesRead;

    // Open source file
    source = open("source.txt", O_RDONLY);

    if (source < 0)
    {
        printf("Error opening source file\n");
        exit(1);
    }

    // Create/Open destination file
    destination = open("destination.txt",
                      O_WRONLY | O_CREAT | O_TRUNC,
                      0644);

    if (destination < 0)
    {
        printf("Error opening destination file\n");
        close(source);
        exit(1);
    }

    // Read from source and write to destination
    while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0)
    {
        write(destination, buffer, bytesRead);
    }

    // Close both files
    close(source);
    close(destination);

    printf("File copied successfully!\n");

    return 0;
}
```

Output:

```
[praveenbusala@Praveens-MacBook-Air-2 CLG % gcc week2.c  
[praveenbusala@Praveens-MacBook-Air-2 CLG % ./a.out  
File copied successfully!
```